Abdulwahab Felemban

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Saudi National KAUST, Saudi Arabia

Fourth-year PhD candidate specializing in multimodal large language models, vision-language models, and vision-language-action systems for autonomous driving, motion prediction, and reasoning-aware planning, with first-author work on iMotion-VLM and iMotion-LLM.

Research internship goal: Seeking a publication-oriented research internship in multimodal autonomy research, with strongest fit in perception, planning, world models, embodied AI, and safety-aware evaluation. Available immediately for Summer/Fall 2026.

RESEARCH INTERESTS

- Multimodal LLMs, VLMs, and VLAs for autonomous driving perception, planning, and generation
- Motion prediction, trajectory generation, world models, and structured reasoning in dynamic scenes
- Safety-aware instruction tuning and evaluation for autonomy systems
- Large-scale autonomous-driving data curation, benchmarking, and closed-loop analysis

SKILLS

Multimodal / VLM / VLA: Qwen-VL, LLaVA, CLIP/SigLIP, instruction tuning (SFT/DPO/GRPO), prompt/data design for multimodal reasoning.

Autonomous Driving / Motion: BEV and trajectory representations, perception-prediction-planning pipelines, open/closed-loop evaluation, Waymo/nuScenes/nuPlan/Bench2Drive/NeuroNCAP workflows.

Training & Systems: PyTorch, multi-GPU training, Slurm job scheduling, vLLM inference, large-scale preprocessing and experiment management.

Programming: Python, OpenCV, MATLAB, R, TensorFlow (basic), C++ (basic).

Deep Learning: Vision transformers, temporal and generative models, end-to-end AD systems, graph deep learning.

Additional Skills: Data curation, metadata cleaning, parallel processing, digital signal/image processing, entrepreneurship.

EDUCATION

PhD **King Abdullah University of Science and Technology (KAUST)**, Electrical & Computer Engineering, Thuwal, Saudi Arabia — 01/2022 - 06/2027

- **GPA:** 3.79/4.0
- **Advisor:** Prof. Mohamed Elhoseiny (`mohamed.elhoseiny@kaust.edu.sa`)
- **Focus:** Deep Learning, Motion Forecasting, and Large Language Models
- **Current Research:** AD with Reasoning; Coral Reef Identification Benchmark

MS **King Abdullah University of Science and Technology (KAUST)**, Electrical & Computer Engineering, Thuwal, Saudi Arabia — 01/2020 - 01/2022

- **GPA:** 3.83/4.0
- **Advisor:** Prof. Tareq Al-Naffouri (`tareq.alnaffouri@kaust.edu.sa`)
- **Focus:** Deep Learning and Human Action Prediction

BS **King Fahd University of Petroleum and Minerals (KFUPM)**, Electrical Engineering, Dhahran, Saudi Arabia — 08/2014 - 12/2019

- **GPA:** 3.43/4.0
- **Concentration:** Signal Processing and Wireless Communication
- **Senior Project:** Real-Time AI Surveillance System
- **Junior Project:** Heatstroke Flexible Sensors

- **Georgia Institute of Technology**, Exchange Student, Atlanta, GA, USA — 08/2018 - 12/2018

OTHER ACADEMIC CONTRIBUTIONS

Reviewer: WACV 2026, ICCV2025 Workshop Proceedings, ICCV 2025 Workshop CV4E, ACML 2024, SIGGRAPH Asia 2024.

Workshop Organizer: CVPR 2024 – C3DV: 2nd Workshop on Compositional 3D Vision. 06/2024

Contributed to the KFUPM DSP Lab Manual: OpenCV Face-Recognition Experiment. 07/2021

AWARDS & FUNDING

CEMSE Dean's List Award Academic Year 21/22 (awarded to the top-20% students)
$1M in Research Funding: NEOM Ocean Science and Solutions (OSSARI) Research Fund ($600k, 2024); KAUST Translational Fund ($450k, 2022).
Projects Awards: Start Smart by Community Jameel Foundation and Bab Rizq Jameel (1st place, $9k, EcoStream Project, 2024), Digital Innovation Award by the Saudi Ministry of Communications and Information Technology (1st place Smart-Tap project, $21k, 2022), KFUPM Undergraduate Senior-Project Award (1st place, AI Surveillance Project, 2019), SSI Internship Poster Competition (2nd place, Indoor localization project, 2019).

EXPERIENCE

KAUST Academy 1-Week AI specialization Instructor: Taught undergraduate students fundamentals of AI (ML, CV, neural networks, Python). 01/2025

KAUST Saudi Summer Internship (SSI): Worked on improving ultrasonic indoor localization by filtering out non-line-of-sight signals with machine learning. 05/2019 - 08/2019

KFUPM IEEE Student Branch Publicity Chair: Organized various events and seminars, and promoted IEEE branch activities to KFUPM students. 01/2018 - 01/2019

PUBLICATIONS

- **Felemban, A.**, Hroub, N., Ding, J., Fu, Y., Hammoud, H., Bezerra, L., Khan, F., Zhang, W., Elhoseiny, M. *iMotion-VLM: Safety-Aligned Vision-Language Models for End-to-End Autonomous Driving* [Submitted, ECCV 2026]
- **Felemban, A.**, Hroub, N., Ding, J., Bakr, E.M., Shen, X., Mohamed, A., Elhoseiny, M. *iMotion-LLM: Motion Prediction Instruction Tuning* [WACV 2026]
- Gou, C., **Felemban, A.**, Khan, F.F., Zhu, D., Cai, J., Rezatofighi, H., Elhoseiny, M. *How Well Can Vision Language Models See Image Details?* [ArXiv:2408.03940, 2024]
- **Felemban, A.***, Battach, Y.* (Equal contribution), Khan, F. F., Radwan, Y. A., Li, X., Marchese, F., Beery, S., Jones, B. H., Benzoni, F., Elhoseiny, M. *ReefNet: A Large-Scale, Taxonomically Enriched Dataset and Benchmark for Hard Coral Classification* [Submitted, ECCV 2026, ArXiv:2510.16822, 2025]
- Khan, F. F., Radwan, Y., Bakr, E.M., **Felemban, A.**, Mir, A., Michiels, N. K., Temple, A. J., Berumen, M. L., Elhoseiny, M. *FishNet++: Analyzing the capabilities of Multimodal Large Language Models in marine biology* [ArXiv:2509.25564, 2025]
- Al-Naffouri, T.Y., Masood, M., **Felemban, A.**, Aljoud, M.F., Bader, A., Alkhodary, M.T., Ahmed, T.B.K. *Real-Time Control of Mechanical Valves Using Computer Vision and Machine Learning* [US Patent App. 18/021,241, 2023]
- **Felemban, A.** *Deep Learning Action Anticipation for Real-Time Control of Water Valves: Wudu Use Case* [MS Thesis, 2021]
- AlSharif, M.H., Ahmed, M., **Felemban, A.**, Zayat, A., Muqaibel, A., Masood, M., Al-Naffouri, T.Y. *Robust 2D Indoor Positioning Algorithm in the Presence of Non-Line-of-Sight Signals* [EUSIPCO 2020]

SELECTED PROJECTS

iMotion-VLM: End-to-End Vision-Language Autonomous Driving System

- Built a reasoning-centric end-to-end planning system using multimodal LLMs for autonomous driving.
- Designed a multi-step reasoning and high-level plan data generation pipeline on nuScenes and safety-focused evaluation settings.

iMotion-LLM, InstructNuPlan / InstructWaymo Datasets

- Developed a safety- and feasibility-aligned data generation pipeline for instruction-following in trajectory prediction using vectorized BEV representations.
- Trained and evaluated instruction adherence and safety alignment by coupling LLMs with traditional trajectory prediction modules.

ReefNet (925K Annotations): Large-Scale Dataset and Benchmark

- Built a 925K-annotation benchmark for hard coral identification in collaboration with marine ecologists.
- Benchmarked ViT/BEiT/BioCLIP/CLIP/SigLIP models and designed a rigorous data verification and quality-control pipeline.

Smart-Tap: Fully Integrated AI System for Water Valve Control

- Built a proof-of-concept prototype with real-time inference for future action anticipation.
- Collected and annotated data with pose estimation, and trained spatio-temporal graph convolutional networks.